

Chemistry 30 — Unit II

Thermochemical Changes

Lesson 4 – Hess' Law

PART A — Why Hess' Law Exists: Energy as a State Function

Big Idea

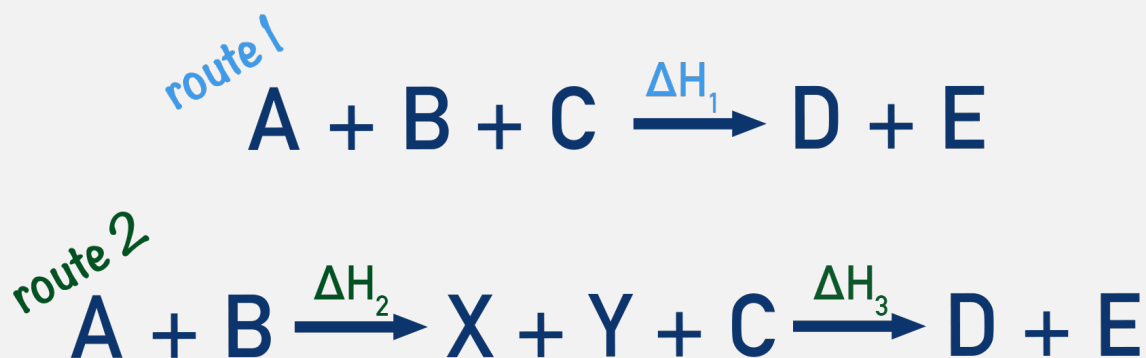
Before Hess' Law can make sense as a calculation tool, it must first make sense as an idea. Hess' Law exists because energy behaves in a very specific and predictable way in chemical reactions. This behaviour is not an arbitrary rule invented by chemists—it is a direct consequence of how energy is conserved.

In earlier lessons, you learned that enthalpy (H) tracks energy changes for systems operating at constant pressure. You also learned that the sign of ΔH tells us whether energy is absorbed by or released from the system. What may not yet be obvious is that enthalpy depends only on where a system starts and where it ends—not on how it gets there.

This idea is known as path independence. It means that no matter how many steps a reaction takes, the total energy change will always be the same as long as the initial reactants and final products are identical. Hess' Law is simply a mathematical way of applying this principle.

Students often assume that breaking a reaction into multiple steps somehow changes the total energy involved. This is a very common misconception. While intermediate steps may involve energy being absorbed and released at different points, those intermediate energy changes cancel out when the full process is considered.

Think of hiking up a mountain. You can walk straight up, or you can follow a winding trail with many turns. You may feel different levels of effort at different moments, but the total change in elevation from start to finish is the same. Enthalpy behaves in exactly the same way.



Enthalpy change for route 1 is the same as the enthalpy change for route 2

$$\Delta H_1 = \Delta H_2 + \Delta H_3$$

Because energy is conserved, chemists can calculate the enthalpy change of reactions that are difficult or impossible to measure directly. By combining reactions with known enthalpy values, we can determine the ΔH for a target reaction without ever performing it experimentally. This is the powerful idea that Hess' Law formalizes.

Part B: Path Independence & Why Hess' Law Works

Big Idea

Before Hess' Law can make sense mathematically, it must make sense conceptually. This section exists to rebuild the idea that energy changes in chemistry depend only on where you start and where you finish—not on the route taken in between.

Students often confuse reaction **pathway** with reaction **rate**. These are not the same. A reaction pathway describes the sequence of steps or intermediate reactions that occur, while the reaction rate describes how fast those steps happen. Hess' Law is concerned only with energy accounting, not speed.

Imagine hiking up a mountain. You can take a steep trail, a winding path, or a road that loops around the mountain. Regardless of which route you take, the change in elevation between the base and the summit is the same. Energy behaves the same way. Enthalpy is a state function—it depends only on the initial and final states of a system.

This is why Hess' Law works. If the total change in enthalpy depends only on the starting materials and the final products, then we are free to break a reaction into steps, reverse steps, or scale steps, as long as everything adds up to the same overall reaction.

A common misconception is that changing the reaction pathway changes ΔH . This is false. Catalysts, for example, change the pathway and lower activation energy, but they do not change the enthalpy difference between reactants and products. The system still starts and ends at the same energy levels.

This idea—path independence—is the conceptual backbone of Hess' Law. Without it, combining thermochemical equations would be meaningless. With it, energy changes become predictable, reliable, and calculable.

Part C: Standard Enthalpy Changes of Formation

Big Idea

To understand Hess' Law fully, we must first understand why chemists rely so heavily on standard enthalpy changes of formation. These values exist because measuring the energy change of every possible chemical reaction directly is neither practical nor efficient. Instead, chemists established a common reference framework that allows energy changes to be calculated indirectly, but reliably.

The standard enthalpy of formation, written as ΔH°_f , is defined as the enthalpy change that occurs when exactly one mole of a compound is formed from its elements in their most stable standard states at standard conditions (25°C and 100 kPa). This definition is very precise, and every word matters.

Students often overlook the importance of the phrase “one mole.” If more or less than one mole is formed, the value is no longer a standard enthalpy of formation. Likewise, if the reactants are not elements in their standard states, the value does not qualify as ΔH°_f .

A crucial consequence of this definition is that the standard enthalpy of formation of any element in its standard state is defined as zero. This is not because no energy exists, but because a reference point is required. By defining elemental standard states as zero, all other enthalpy changes can be measured relative to that baseline.

For example, the standard enthalpy of formation of $\text{O}_2(\text{g})$, $\text{H}_2(\text{g})$, and $\text{C}(\text{s, graphite})$ is zero. However, the standard enthalpy of formation of $\text{CO}_2(\text{g})$ is -394 kJ/mol , indicating that energy is released when carbon and oxygen combine to form carbon dioxide under standard conditions.

This reference system is what makes Hess' Law workable. Because enthalpy is a state function, the total energy change for a reaction depends only on the initial and final states, not on the pathway taken. Formation enthalpies allow us to define those states numerically.

In practical terms, calculating the enthalpy change of a reaction using formation enthalpies involves subtracting the total formation enthalpy of the reactants from the total formation enthalpy of the products. This method avoids tracking every intermediate step explicitly.

Students sometimes confuse formation enthalpies with combustion enthalpies. While both involve energy changes, formation enthalpies are strictly tied to element-to-compound formation, whereas combustion enthalpies involve reaction with oxygen. Keeping these concepts distinct is essential for correct application.

By the end of this section, you should see standard enthalpy changes of formation not as isolated numbers to memorize, but as a carefully constructed energy accounting system that allows complex reactions to be analyzed with clarity and confidence.

Part D: Applying Formation Enthalpies to Calculate ΔH

Big Idea

In this section, students learn how Hess' Law is applied using standard enthalpies of formation. Rather than manipulating equations directly, this method relies on a consistent algebraic relationship between reactants and products. Because enthalpy is a state function, the overall energy change depends only on the initial and final states of a reaction, not on the pathway taken.

The standard enthalpy of formation, ΔH°_f , represents the energy change when one mole of a compound forms from its elements in their standard states. When multiple substances react, the total enthalpy change can be determined by comparing the summed formation enthalpies of products and reactants.

This leads to the Hess' Law formula method:

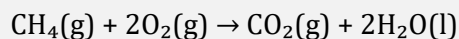
$$\Delta H^{\circ}\text{reaction} = \Sigma n\Delta H^{\circ}\text{f}(\text{products}) - \Sigma n\Delta H^{\circ}\text{f}(\text{reactants})$$

Each coefficient in the balanced chemical equation acts as a multiplier, scaling the contribution of each substance's formation enthalpy.

A common student misconception is assuming the sign of $\Delta H^{\circ}\text{f}$ must always be negative. In reality, the sign reflects whether energy is released or absorbed during formation, and positive values are both valid and meaningful.

Worked Example:

Calculate ΔH° for the combustion of methane:



Using standard enthalpies of formation:

$$\Delta H^{\circ}\text{f CO}_2(\text{g}) = -393.5 \text{ kJ/mol}$$

$$\Delta H^{\circ}\text{f H}_2\text{O}(\text{l}) = -285.8 \text{ kJ/mol}$$

$$\Delta H^{\circ}\text{f CH}_4(\text{g}) = -74.8 \text{ kJ/mol}$$

$$\Delta H^{\circ}\text{f O}_2(\text{g}) = 0 \text{ kJ/mol}$$

$$\Delta H^{\circ}\text{reaction} = [1(-393.5) + 2(-285.8)] - [1(-74.8) + 2(0)] = -890.3 \text{ kJ}$$

This approach is powerful because it avoids trial-and-error equation manipulation and provides a direct, reliable calculation method aligned with Alberta Diploma expectations.

PART E — Algebraic Manipulation of Thermochemical Equations

Big Idea

In this section, you will learn the traditional algebraic method of applying Hess' Law. This approach relies on careful manipulation of thermochemical equations rather than direct use of standard enthalpies of formation. Although it may initially feel procedural, the reasoning behind each step is grounded firmly in energy conservation and the state-function nature of enthalpy.

Hess' Law states that the total enthalpy change for a chemical reaction depends only on the initial and final states of the system, not on the pathway taken. Because of this, reactions can be broken into steps, rearranged, reversed, or scaled, as long as the overall chemical equation is preserved. The enthalpy changes associated with each step combine algebraically to give the total ΔH .

The first skill required is reversing thermochemical equations. When a reaction is reversed, the direction of energy flow also reverses. This means the sign of ΔH must

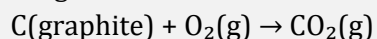
change. A common student error is reversing the chemical equation but forgetting to reverse the sign of the enthalpy change. Always remember: if reactants and products swap places, the sign of ΔH must flip.

The second essential skill is scaling equations. Enthalpy is an extensive quantity, meaning it depends on the amount of substance reacting. If all coefficients in a reaction are multiplied by a factor, the enthalpy change must be multiplied by the same factor. Students often incorrectly assume ΔH stays constant; in fact, ΔH scales directly with the reaction quantity.

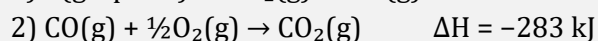
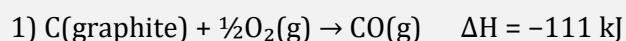
Once equations are correctly oriented and scaled, they can be added together. Any species that appears on both sides of the combined equations cancels, just as in algebra. The remaining species form the target net reaction. The enthalpy changes are then summed to find the overall ΔH .

Worked Example:

Target reaction:



Given:



Adding the equations cancels $\text{CO}(\text{g})$, producing the target reaction. Adding the enthalpy values:

$$\Delta H = -111 \text{ kJ} + (-283 \text{ kJ}) = -394 \text{ kJ}$$

This confirms that the enthalpy change for the formation of CO_2 from graphite is -394 kJ .

Throughout this process, careful attention to signs, coefficients, and cancellation is essential. Hess' Law problems reward precision and logical reasoning more than speed. Practicing this method builds strong foundational skills that transfer directly to diploma-level thermochemistry questions.

Part F: Comparing Hess' Law Methods

Big Idea

In this final section of the Hess' Law lesson, we step back from calculations and focus on strategy. By this point, you have learned two distinct but equivalent ways

of determining enthalpy changes: using standard enthalpies of formation, and using algebraic manipulation of thermochemical equations. Both methods are grounded in the same physical principle—enthalpy is path independent—but each has advantages depending on the information provided.

The formation-enthalpy method is often more direct. When standard enthalpies of formation are available, the overall enthalpy change can be calculated using a single equation that subtracts the total enthalpy of the reactants from that of the products. This approach minimizes algebraic complexity and reduces the chance of sign errors. It is particularly effective when reactions involve common substances with well-tabulated formation values.

The algebraic method, by contrast, emphasizes reasoning with equations. Rather than relying on tabulated values, you construct the desired reaction by reversing and scaling known reactions. This method is more flexible, especially when formation enthalpies are unavailable or when reactions are given in unusual forms. However, it requires careful tracking of coefficients and enthalpy signs at each step.

A common misconception is that one method is ‘more correct’ than the other. In reality, both must give the same result if applied correctly. Differences in student answers almost always arise from procedural errors—forgetting to reverse the sign of ΔH when reversing an equation, or failing to scale ΔH when coefficients change.

On diploma-style questions, the method you choose should be strategic. If formation enthalpies are given or easily accessible, that method is usually faster and safer. If you are given multiple reactions with associated ΔH values, the algebraic method may be unavoidable. Strong students recognize which approach is most efficient before writing a single line of math.

Ultimately, both approaches reinforce the same conceptual truth: energy accounting in chemistry is consistent and conserved. Hess’ Law is not a shortcut—it is a reflection of how nature tracks energy changes regardless of the route taken.

Chemistry 30 – Lesson 4: Hess' Law Practice

This practice set is designed to reinforce your understanding of Hess' Law. Work carefully through each question, showing all reasoning and algebraic steps. A complete worked-solution section follows this practice so you can check not just your answers, but your reasoning.

Part A: Conceptual Understanding

1. Explain, in your own words, why enthalpy is described as a state function.
2. Explain why Hess' Law allows us to add and subtract equations to find ΔH for a net reaction.

Part B: Algebraic Manipulation of Equations

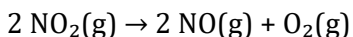
3. Given the following thermochemical equations, determine the enthalpy change for the target reaction:



Target: $A \rightarrow C$

Part C: Formation Enthalpy Method

4. Calculate the enthalpy change for the following reaction using standard enthalpies of formation:



Worked Solutions – Lesson 4 Practice

Solution 1

Enthalpy is a state function because it depends only on the initial and final states of a system, not on the path taken between them. This means that regardless of the sequence of steps, the total energy change will be the same.

Solution 2

Hess' Law works because reactions can be combined algebraically without changing the overall energy difference between reactants and products. Since ΔH depends only on start and end states, intermediate steps cancel out.

Solution 3

To obtain $A \rightarrow C$, the given equations are added directly. The ΔH values are also added:

$$\Delta H = -120 \text{ kJ} + 45 \text{ kJ} = -75 \text{ kJ}$$

Solution 4

Using formation enthalpies, ΔH is calculated by subtracting the sum of the reactants' formation enthalpies from the sum of the products' formation enthalpies.

Diploma-Style Practice: Section B–C Extended Response

This practice set is designed to mirror the structure, depth, and reasoning expected on the Alberta Chemistry 30 Diploma Examination. Students are expected to communicate clearly, justify each step, and show algebraic reasoning explicitly.

Section B – Written Response (Reasoning & Explanation)

1. Explain why Hess' Law is valid even when a reaction occurs through multiple steps.

Response:

2. Describe the role of standard enthalpies of formation in calculating ΔH for a reaction.

Response:

3. Explain why reversing a thermochemical equation changes the sign of ΔH .

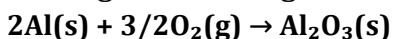
Response:

4. A student claims that catalysts change the overall ΔH of a reaction. Explain why this claim is incorrect.

Response:

Section C – Calculation & Analysis (Multi-Step)

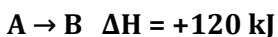
5. Using the following formation enthalpies, calculate ΔH for the reaction:



$$\Delta H_f^\circ[\text{Al}_2\text{O}_3\text{(s)}] = -1676 \text{ kJ/mol}$$

Show all work and reasoning:

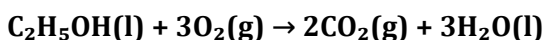
6. Given the following thermochemical equations:



Determine ΔH for $\text{A} \rightarrow \text{C}$ using Hess' Law.

Show all work and reasoning:

7. Calculate ΔH for the combustion of ethanol using formation enthalpies.



Show all work and reasoning:

8. A reaction occurs in three steps. Two steps are endothermic and one is exothermic. Explain how the overall ΔH can still be negative.

Fully Worked Solutions

The following solutions model the depth of explanation expected from high-achieving students on the Chemistry 30 Diploma Examination.

Solution 1:

Hess' Law is valid because enthalpy is a state function. This means that the total enthalpy change depends only on the initial and final states of the system, not the pathway taken between them. Even if a reaction proceeds through multiple intermediate steps, the total energy change must be conserved.

Solution 2:

Standard enthalpies of formation allow chemists to calculate reaction enthalpies by comparing the energy of products and reactants relative to their elements in standard states. This provides a consistent reference point.

Solution 3:

Reversing a thermochemical equation reverses the direction of energy flow. Energy that was released must now be absorbed, so the sign of ΔH changes.

Solution 4:

Catalysts lower activation energy but do not alter the energies of reactants or products. Since ΔH depends only on these energy levels, catalysts cannot change the overall enthalpy change.

Solution 5:

$$\Delta H = \Sigma \Delta H_f^\circ(\text{products}) - \Sigma \Delta H_f^\circ(\text{reactants})$$

$$\Delta H = -1676 \text{ kJ} - 0 = -1676 \text{ kJ}$$

The reaction is strongly exothermic.

Solution 6:

$$\Delta H(A \rightarrow C) = \Delta H(A \rightarrow B) + \Delta H(B \rightarrow C)$$

$$\Delta H = +120 \text{ kJ} + (-75 \text{ kJ}) = +45 \text{ kJ}$$

Solution 7:

Students calculate ΔH by summing formation enthalpies of products and subtracting those of reactants, accounting for coefficients.

Solution 8:

Even with multiple endothermic steps, a sufficiently large exothermic step can result in a net release of energy.